Cyber Security (Section- E4)
Assignment-02 (Fall 2026)

Submission to be done on portal

Due Date: Wednesday, November 18, 2026, before 10:00 AM

- **Use Assignment pages and scan your assignment. (Hand written Only)**
- **Upload .pdf file on portal**
- **[Note: Title page must have Student Name, Registration Number, Section, and Date of Submission.**
- **Don't use register pages**
- **Submissions will be accepted both through the portal and must be submitted in hard copy format in class.**

Question 1

An employee receives an email that appears to come from the HR department. It asks for their login username and password so HR can "update personal information" before the end of the day.

1. List **four warning signs** in this email that suggest it is a phishing attempt (think about sender address, urgency, links, and the request itself).
2. Write, step by step, what the employee should do — and what they should **not** do.
3. What should the IT/security team do after the email is reported, so that other employees are protected too?
4. Why do attackers pretend to be HR or the CEO instead of an unknown person? Name the psychological trick being used.

Question 2

During an audit, you find that a group of employees uses weak passwords such as `123456`, `company2026` and their own names to log in to critical company applications.

1. Explain **two attacks** that become easy because of weak passwords (e.g., brute force, dictionary attack, credential stuffing).
2. Propose a solution in three layers: **Technical** (policy and tools), **Human** (training and awareness), and **Monitoring** (how you will check it keeps working).
3. Write a sample password policy of **5 rules** for this company.
4. Even a strong password can be stolen by phishing. How does **Multi-Factor Authentication (MFA)** protect the account in that case?

Question 3

A smart-home security company discovers that one of its IoT cameras can be accessed remotely **without any authentication**, allowing attackers to watch private footage from people's homes.

1. Explain the main security risks of IoT devices (e.g., default credentials, no updates, insecure network services, unencrypted data, weak privacy controls).
2. Suggest **security practices** to fix this vulnerability for cameras already sold (short term) and **policies** to prevent it in future products (long term, e.g., secure by design, firmware update process, vulnerability disclosure program).
3. What should the company tell its customers, and why does honest communication matter?

Question 4

Explain each of the following attacks in **one short paragraph** (what happened, when, and who was affected):

1. **WannaCry** ransomware (2017)
2. **Stuxnet** (discovered 2010)
3. **Slowloris** denial-of-service attack
4. **SolarWinds** supply chain attack (2020)
5. **Mirai** botnet attack (2016)

Then copy and complete this table by hand:

Attack	Threat / Attack type	Attack vector (how it entered and spread)	Impact	One lesson for defenders
WannaCry				
Stuxnet				
Slowloris				
SolarWinds				
Mirai				

Question 5

1. Define **system vulnerability**.
2. Explain how a vulnerability differs from a **threat** and an **attack**. Use a real-life analogy (e.g., an unlocked door, a thief in the area, and the actual break-in).
3. Give **four examples** of common system vulnerabilities discussed in the literature, such as buffer overflow, SQL injection, misconfiguration, unpatched software, and weak authentication. For each, write one line on how it can be exploited.
4. Write the relationship: **Risk = Threat × Vulnerability × Impact**, and explain it with one example.

Question 6

1. Define **payload** in the context of malware. How is the payload different from the part of the malware that spreads it?
2. Compare **benign**, **malicious** and **covert** payloads. Copy and fill this table:

Payload type	What it does	Visible to the user?	Example
Benign			
Malicious			
Covert			

Question 7

A university computer lab reports that machines have become **extremely slow**, **unwanted pop-ups** appear, and **students' login information is being stolen**.

1. Identify **two types of malware** that could be responsible. Link each symptom to the malware type that best explains it.
2. Describe the **infection mechanism** of each (how it got onto the lab machines — e.g., infected USB, malicious download, bundled freeware, phishing link).
3. Explain the **impact** of each on the system and on the users.

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4. As the lab in-charge, list **five steps** you would take to clean the lab and prevent this from happening again.

Question 8

Consider this case: *A ransomware attack begins with a phishing email, installs a backdoor, spreads laterally using a Windows SMB vulnerability, encrypts files, and demands payment.*

1. Explain each stage of the **Malware Attack Lifecycle** (reconnaissance, delivery, exploitation/installation, command and control, lateral movement, actions on objectives).
2. Map every event from the case to the correct stage using a table:

Lifecycle stage	Event in the case	One defence that could stop it here

3. Draw the lifecycle as a flow diagram with arrows.

Question 9

1. Differentiate between a **system error**, a **software bug**, and a **security vulnerability**, with one example of each.
2. Can all system errors be considered vulnerabilities? Justify your answer using textbook concepts.

Question 10

Your organization wants to strengthen its perimeter defences. Explain how the combined use of a **Firewall**, an **Intrusion Detection System (IDS)** and an **Intrusion Prevention System (IPS)** provides **layered security (defence in depth)**.

1. Explain what each component does and where it sits in the network.
2. Copy and complete this comparison table:

Component	What it checks	Action it takes	Benefits	Limitations
Firewall				
IDS				
IPS				

3. Draw a labelled network diagram: **Internet → Firewall → IPS → internal network**, with the IDS monitoring a copy of the traffic.
4. Describe how all three work together when an attacker sends a malicious packet that the firewall allows (e.g., an exploit hidden in normal web traffic on port 80/443).

Research Questions: These questions are not required to be handwritten. You may type your answers, print them, and attach the printed pages to your handwritten assignment.

A **Security Operations Centre (SOC)** is the team, processes and technology that monitor an organization 24/7, detect attacks, and respond to incidents. Almost every concept in this assignment like phishing , breach response , access monitoring , IoT threats , the attack lifecycle and IDS alerts is handled inside a SOC.

Part (a)

1. Define a SOC and explain its main functions: monitoring, detection, triage, incident response, and threat hunting.

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2. Describe the SOC team roles: **Tier 1** (alert triage), **Tier 2** (incident investigation), **Tier 3** (threat hunting / expert), and the SOC manager.

Part (b)

Choose **three** questions from this assignment (Q1–Q9). For each, explain in 3–4 lines how a SOC would detect and respond to that situation.

Part (c)

Find and read a **minimum of 5 research studies published in 2025 or later** about SOC challenges or improvements. Suitable topics include:

- Alert fatigue and false positives in SOCs
- AI / machine learning / large language models (LLMs) for alert triage and incident response
- SOAR and security automation
- SOC analyst burnout, skills shortage and human factors
- SOCs for small and medium enterprises (SMEs) and managed security services
- Threat intelligence integration and SOC performance metrics

Where to search: **Google Scholar, IEEE Xplore, ACM Digital Library, ScienceDirect, SpringerLink, arXiv**. Useful keywords: *"security operations center" challenges*, *SOC alert fatigue*, *LLM incident response*, *SOAR automation*, *SOC SME*.

For each study, copy and fill this table by hand:

#	Title, authors, year, venue	Problem studied	Method used	Key findings	Limitation / gap
1					
2					
3					
4					
5					

Then write a short **synthesis** (half a page): what problems appear again and again across the studies?

Part (d)

Based on the gaps you found, propose **at least three future directions** for solving SOC problems. For each direction, write:

1. **The problem** it addresses (linked to a gap from Part c).
2. **Your proposed solution** (e.g., an AI assistant that summarises and ranks alerts, an affordable automated SOC for small businesses, better playbooks, analyst training through simulation).
3. **How you as a computer science student could contribute** — a project, tool, dataset, or study you could build or carry out.
4. **One challenge or risk** of your idea (e.g., AI making wrong decisions, privacy, cost, trust).

End with a **reference list in APA or IEEE style**. Every study in your table must appear in it. Fake or unverifiable references get zero marks for Part (c).

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